

Probability of Disease

In symbols:

$$P(H|E) = [P(E|H) \times P(H)] / [P(E|H) \times P(H) + P(E|\neg H) \times P(\neg H)]$$

In written language:

The probability of the hypothesis given the evidence equals the probability of the evidence given the hypothesis times the probability of the hypothesis, all divided by the quantity of the probability of the evidence given the hypothesis times the probability of the hypothesis, plus the probability of the evidence given not the hypothesis times the probability of not the hypothesis.